A SECOND VERSION OF THE OLSSON-LASZLO BOUNDEDNESS LEMMA FOR A MORPHISM OF RINGED SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the derived limit presentation, the eventual constancy of the cohomology groups on a cofinal system of coverings, the identification of the cohomology of the derived limit and the final truncation comparison are all written out, with every cited input and the dimension bound named.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans: the author's statement environment carries, inside a 'reference' environment at source lines 5877-5880, a provenance remark citing a published lemma of Olsson and Laszlo, and that remark is not part of the mathematical statement; it is therefore not delivered, and the counted statement is split into the two spans 5875-5876 and 5881-5896. No citation was supplied, invented or reconstructed, and no other text of the author was dropped.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The two Olsson-Laszlo situations. Same chapter, Situations situation-olsson-laszlo and situation-olsson-laszlo-prime. Supply the hypotheses referred to by assumptions (1) and (2).
- Unbounded complexes are derived limits of their truncations. Same chapter, Lemma lemma-olsson-laszlo. Writes the complex as the derived limit of its canonical truncations.
- The derived pushforward commutes with derived limits, and cohomology of a derived limit. Same chapter, Lemmas lemma-Rf-commutes-with-Rlim and lemma-RGamma-commutes-with-Rlim. Move the derived limit through the pushforward and identify the values of the resulting cohomology sheaves.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

- Constancy of a derived limit of a constant system. Same chapter, Lemma lemma-cohomology-derived-limit-injective. Turns the constancy of the system of cohomology groups into the asserted isomorphism.

Source reference. Original lemma 0D6S: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5707–5707. Navigation pointer invoked at proof line 5826; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0A07: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5150–5150. Navigation pointer invoked at proof line 5827; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6K: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5126–5126. Navigation pointer invoked at proof line 5791; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6L: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5279–5279. Navigation pointer invoked at proof line 5759; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 015J: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5742–5742. Navigation pointer invoked at proof line 5742; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 08J5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5948–5948. Navigation pointer invoked at proof line 5948; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Situation 0.1. Let $\mathcal{C}$ be a site. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$. Let $\mathcal{A} \subset \mathit{Mod}(\mathcal{O})$ be a weak Serre subcategory. We assume the following is true: there exists a subset $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ such that

(1) every object of $\mathcal{C}$ has a covering whose members are in $\mathcal{B}$, and
(2) for every $V \in \mathcal{B}$ there exists an integer d_V and a cofinal system Cov_V of coverings of V such that

$$H^p(V_i, \mathcal{F}) = 0 \text{ for } \{V_i \to V\} \in \mathrm{Cov}_V,\ p > d_V,\ \text{and } \mathcal{F} \in \mathrm{Ob}(\mathcal{A})$$

Situation 0.2. Let $f : (\mathcal{C}, \mathcal{O}) \to (\mathcal{C}', \mathcal{O}')$ be a morphism of ringed sites. Let $u : \mathcal{C}' \to \mathcal{C}$ be the corresponding continuous functor of sites. Let $\mathcal{A} \subset \mathit{Mod}(\mathcal{O})$ be a weak Serre subcategory. We assume the following is true: there exists a subset $\mathcal{B}' \subset \mathrm{Ob}(\mathcal{C}')$ such that

(1) every object of $\mathcal{C}'$ has a covering whose members are in $\mathcal{B}'$, and
(2) for every $V' \in \mathcal{B}'$ there exists an integer $d_{V'}$ and a cofinal system $\mathrm{Cov}_{V'}$ of coverings of V' such that

$$H^p(u(V_i'), \mathcal{F}) = 0 \text{ for } \{V_i' \to V'\} \in \mathrm{Cov}_{V'},\ p > d_{V'},\ \text{and } \mathcal{F} \in \mathrm{Ob}(\mathcal{A})$$

Lemma 0.3. *Let $f : (\mathcal{C}, \mathcal{O}) \to (\mathcal{C}', \mathcal{O}')$ be a morphism of ringed sites. assume moreover there is an integer N such that*

(1) *$\mathcal{C}, \mathcal{O}, \mathcal{A}$ satisfy the assumption of Situation 0.1,*
(2) *$f : (\mathcal{C}, \mathcal{O}) \to (\mathcal{C}', \mathcal{O}')$ and $\mathcal{A}$ satisfy the assumption of Situation 0.2,*
(3) *$R^p f_* \mathcal{F} = 0$ for $p > N$ and $\mathcal{F} \in \mathrm{Ob}(\mathcal{A})$,*

Then for K in $D_{\mathcal{A}}(\mathcal{O})$ the map $H^j(Rf_*K) \to H^j(Rf_*(\tau_{\geq -n}K))$ is an isomorphism for $j \geq N - n$.

Proof. Let K be in $D_{\mathcal{A}}(\mathcal{O})$. By Lemma 0D6S we have $K = R\lim \tau_{\geq -n}K$. By Lemma 0A07 we have $Rf_*K = R\lim Rf_*(\tau_{\geq -n}K)$. Let $V' \in \mathcal{B}'$ and let $\{V'_i \to V'\}$ be an element of $\mathrm{Cov}_{V'}$. Then we consider

$$H^j(V'_i, Rf_*K) = H^j(u(V'_i), K) \quad \text{and} \quad H^j(V'_i, Rf_*(\tau_{\geq -n}K)) = H^j(u(V'_i), \tau_{\geq -n}K)$$

The assumption in Situation 0.2 implies that the last group is independent of n for n large enough depending on j and $d_{V'}$. Some details omitted. We apply this for j and $j-1$ and via Lemma 0D6K this gives that

$$H^j(V'_i, Rf_*K) = \lim H^j(V'_i, Rf_*(\tau_{\geq -n}K))$$

and the system on the right is constant for n larger than a constant depending only on $d_{V'}$ and j. Thus Lemma 0D6L implies that

$$H^j(Rf_*K)(V') \longrightarrow \left(\lim H^j(Rf_*(\tau_{\geq -n}K))\right)(V')$$

is injective. Since the elements $V' \in \mathcal{B}'$ cover every object of $\mathcal{C}'$ we conclude that the map $H^j(Rf_*K) \to \lim H^j(Rf_*(\tau_{\geq -n}K))$ is injective. The spectral sequence

$$E_2^{p,q} = R^p f_* H^q(\tau_{\geq -n}K)$$

converging to $H^{p+q}(Rf_*(\tau_{\geq -n}K))$ (Derived Categories, Lemma 015J) and assumption (3) show that $H^j(Rf_*(\tau_{\geq -n}K))$ is constant for $n \geq N - j$. Hence $H^j(Rf_*K) \to H^j(Rf_*(\tau_{\geq -n}K))$ is injective for $j \geq N - n$.

Thus we proved the lemma with "isomorphism" in the last line of the lemma replaced by "injective". However, now choose j and n with $j \geq N - n$. Then consider the distinguished triangle

$$\tau_{\leq -n-1}K \to K \to \tau_{\geq -n}K \to (\tau_{\leq -n-1}K)[1]$$

See Derived Categories, Remark 08J5. Since $\tau_{\geq -n}\tau_{\leq -n-1}K = 0$, the injectivity already proven for $\tau_{-n-1}K$ implies

$$0 = H^j(Rf_*(\tau_{\leq -n-1}K)) = H^{j+1}(Rf_*(\tau_{\leq -n-1}K)) = H^{j+2}(Rf_*(\tau_{\leq -n-1}K)) = \ldots$$

By the long exact cohomology sequence associated to the distinguished triangle

$$Rf_*(\tau_{\leq -n-1}K) \to Rf_*K \to Rf_*(\tau_{\geq -n}K) \to Rf_*(\tau_{\leq -n-1}K)[1]$$

this implies that $H^j(Rf_*K) \to H^j(Rf_*(\tau_{\geq -n}K))$ is an isomorphism. □